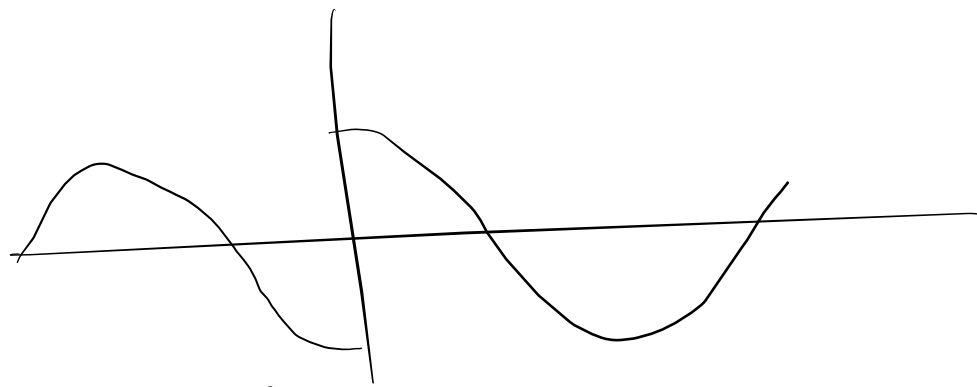


Differentiability:  $f'(x) = \lim_{h \rightarrow 0} \frac{f(x+h) - f(x)}{h}$  exists  $\forall x$ .

$f'(a) = \lim_{h \rightarrow 0} \frac{f(a+h) - f(a)}{h}$  exists, then

$f(x)$  is differentiable at point  $x=a$ .

1. (i)  $f(x) = \begin{cases} \cos x, & x \geq 0 \\ \boxed{-\cos x}, & x < 0, \end{cases}$  at  $\boxed{x=0}$ .



$$f(0) = \cos 0 = 1.$$

Differentiable  $\Rightarrow$  Continuous  $\Rightarrow$  limit exists

~~$\nRightarrow$~~

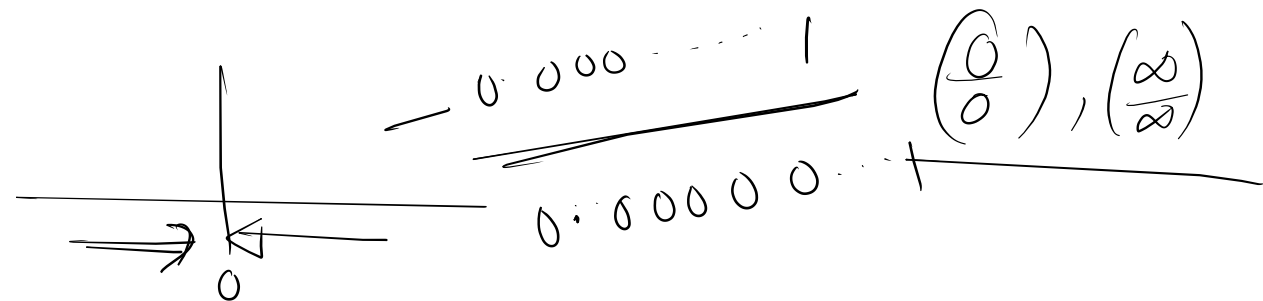
~~$\nRightarrow$~~  exists

$$\underline{\underline{L.H.D.}} = \lim_{h \rightarrow 0^-} \frac{f(0+h) - f(0)}{h}$$

$$= \lim_{h \rightarrow 0^-} \frac{f(h) - f(0)}{h}$$

$$= \lim_{h \rightarrow 0^-} \frac{-\cosh h - 1}{h} = \lim_{h \rightarrow 0^-} \frac{-(1 + \cosh h)}{h}$$

$$= \lim_{h \rightarrow 0^-} \frac{-2\cosh^2 \frac{h}{2}}{h} = \underline{\underline{-\infty}}$$



$$f(h) = -\cosh h$$

$$R.H.D. = \lim_{h \rightarrow 0^+} \frac{f(0+h) - f(0)}{h} = \lim_{h \rightarrow 0^+} \frac{\cosh h - 1}{h}$$

$$= \lim_{h \rightarrow 0^+} \frac{-2\sinh^2 \frac{h}{2} \left(\frac{0}{0}\right)}{h} = \lim_{h \rightarrow 0^+} \frac{-2 \cancel{\sinh^2} \frac{h}{2} \cdot \cosh \frac{h}{2} \cdot \cancel{\frac{1}{2}}}{h}$$

$$= \lim_{h \rightarrow 0^+} -\sinh h$$

$$= \lim_{h \rightarrow 0} \sinh = \textcircled{0}.$$

$$\text{L.H.D.} \neq \text{R.H.D.}$$

$$\lim_{h \rightarrow 0} \frac{f(0+h) - f(0)}{h} \text{ does not exist.}$$

$f(x)$  is not differentiable at  $x=0$ .

$$(ii) \quad f(x) = \begin{cases} \frac{x^{\sqrt{x}} \sin\left(\frac{1}{x}\right), & x \neq 0 \\ \underline{\underline{0}}, & \underline{\underline{x=0}}. \end{cases} \quad \text{at } x=0.$$

$$f(0) = 0, \quad f(0+h) = f(h) = h^{\sqrt{h}} \sin\left(\frac{1}{h}\right).$$

$$f'(0) = \lim_{h \rightarrow 0} \frac{f(0+h) - f(0)}{h}$$

$$= \lim_{h \rightarrow 0} \frac{f(h) - f(0)}{h}$$

$$= \lim_{h \rightarrow 0} \frac{\cancel{h^{\sqrt{h}}} \sin\left(\frac{1}{h}\right)}{\cancel{h}}$$

$$= \lim_{h \rightarrow 0} \frac{h^{\sqrt{h}} \sin\left(\frac{1}{h}\right)}{h}$$

$$= \underline{\underline{0}}$$

By Squeeze theorem,

$$-1 \leq \sin\left(\frac{1}{h}\right) \leq 1$$

$$\Rightarrow -h \leq h \sin\left(\frac{1}{h}\right) \leq h$$

$$g(x) \leq f(x) \leq h(x)$$

$$\lim_{x \rightarrow a} g(x) = \lim_{x \rightarrow a} h(x) = l$$

$$\text{then } \lim_{x \rightarrow a} f(x) = l.$$

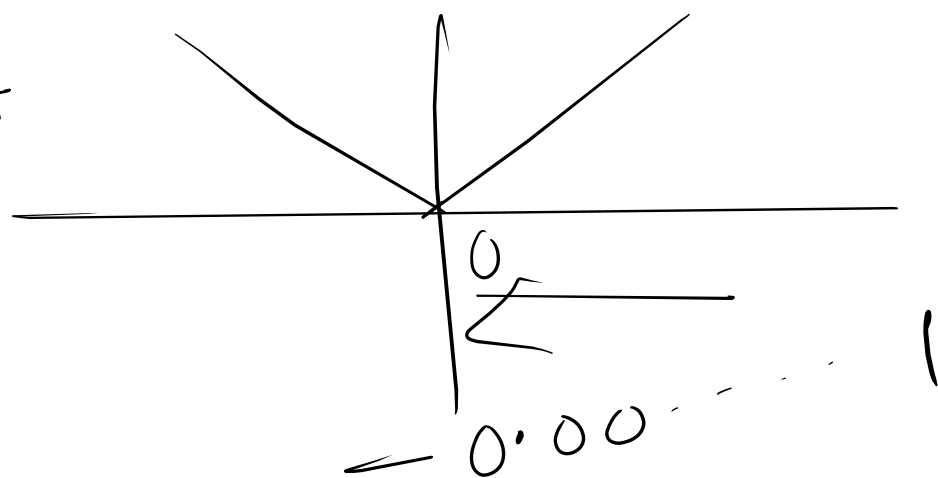
$$\lim_{h \rightarrow 0} -h = \lim_{h \rightarrow 0} h = 0,$$

$$\lim_{h \rightarrow 0} h \sin\left(\frac{1}{h}\right) = 0.$$

$f(x)$  is differentiable at  $x=0$ .

(iii)  $f(x) = |x|$ , at  $x=0$

$$|x| = \begin{cases} x, & x \geq 0 \\ -x, & x < 0 \end{cases}$$



$f(0) = 0$

L.H.D.  $= \lim_{h \rightarrow 0^-} \frac{f(h) - f(0)}{h}$

$f(h) = -h$

$= \lim_{h \rightarrow 0^-} \frac{-h - 0}{h} = -1$

R.H.D.  $= \lim_{h \rightarrow 0^+} \frac{f(h) - f(0)}{h} = \lim_{h \rightarrow 0^+} \frac{h}{h} = 1$

$f(h) = h$

L.H.D.  $\neq$  R.H.D.

~~⊛~~

$$f(x) = |x| + |x-1|, \text{ at } \boxed{x=0, 1}$$

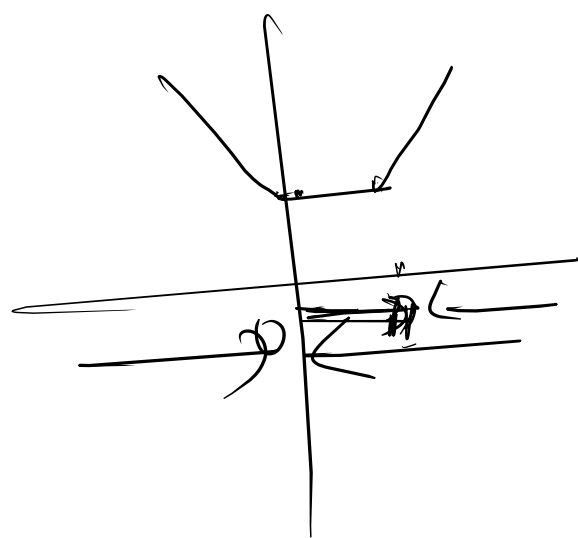
+0.00000-1

$$f(x) = \begin{cases} -2x+1, & \boxed{x < 0} \end{cases}$$

$$y = \begin{cases} -2x+1 \end{cases}$$

$$f(0) = 1$$

$$\begin{cases} 2x-1, & \boxed{0 \leq x < 1} \\ x > 1 \end{cases}$$



$$\text{L.H.D.} = \lim_{h \rightarrow 0^-} \frac{f(h) - f(0)}{h}$$

✓✓

$$= \lim_{h \rightarrow 0^-} \frac{-2h+1-1}{h} = \underline{\underline{-2}}$$

$$R.H.D. = \lim_{h \rightarrow 0^+} \frac{f(h) - f(0)}{h}$$

$$= \lim_{h \rightarrow 0^+} \frac{1-1}{h} = \underline{\underline{0}}$$

$L.H.D. \neq R.H.D.$   $\lim_{h \rightarrow 0} f(x)$  does not exist at  $x=0$ .  
 $f(x)$  is not differentiable at  $x=0$ .

for 1

$$f(1) = 2 \cdot (1-1) = 1$$

$$L.H.D. = \lim_{h \rightarrow 0} \frac{f(1+h) - f(1)}{h}$$

$$= \lim_{h \rightarrow 0} \frac{1-1}{h} = \underline{\underline{0}}$$



$$R.H.D. = \lim_{h \rightarrow 0^+} \frac{f(1+h) - f(1)}{h}$$

$$f(1+h) = 2(h+1) - 1$$

$$= \lim_{h \rightarrow 0^+} \frac{2(h+1) - 1 - 1}{h} = \lim_{h \rightarrow 0^+} \frac{2h + 2 - 2}{h}$$

$$= 2$$

$$L.H.D. \neq R.H.D.$$